oxygen linearly overlap with the sp^3 hybridised orbital of carbon to form a C-O, 'σ' bond and another sp^3 hybridised orbital linearly overlap with 1s orbital of hydrogen to form a O-H 'σ' bond. The remaining two sp^3 hybridised orbitals of oxygen are occupied by two lone pairs of electrons. Due to the lone pair – lone pair repulsion, the C-O-H bond angle in methanol is reduced to $108.9°$ from the regular tetrahedral bond angle of $109.5°$.

Preparation of alcohols:

We have already learnt that the nucleophilic substitution reactions of alkyl halides with dilute alkali, conversion of alkenes to alcohols by hydration and the preparation of alcohols using Grignard reagent in XI standard. These reactions are summarised below.

1. **From Alkyl halides:** Alkyl halides on heating with dilute aqueous NaOH gives alcohols. Primary alkyl halides undergo substitution by SN^2 reaction. Secondary and tertiary alkyl halides usually undergo nucleophilic substitution by SN^1 mechanism.

$$R\text{-}X + NaOH(aq) \xrightarrow{\Delta} R\text{-}OH + NaX$$

If R =t-butyl, the reaction proceeds through the formation of t-butyl carbocation

2. **From alkenes:** Addition of water across the double bond of an alkene in presence of concentrated sulphuric acid gives alcohols. This addition reaction follows Markownikoff's rule.

Example: $CH_3\text{-}CH{=}CH_2 + H_2O \xrightarrow{Conc.H_2SO_4} CH_3\text{-}CH(OH)\text{-}CH_3$

propylene — propan-2-ol

3. **From Grignard reagent:** Nucleophilic addition of Grignard reagent to aldehydes/ketones in presence of dry ether followed by the acid hydrolysis gives alcohols. Formaldehyde gives primary alcohol and other aldehydes give secondary alcohols. Ketones give tertiary alcohols.

Examples

$C_6H_5MgBr + HCHO \xrightarrow{Ether} C_6H_5-\underset{H}{\overset{H}{\underset{|}{\overset{|}{C}}}}-OMgBr \xrightarrow[H^+]{H_2O} C_6H_5CH_2OH + Mg(OH)Br$

Phenyl magnesium bromide — Methanal (Formaldehyde) — Phenyl methanol

$CH_3CH_2MgBr + CH_3CHO \xrightarrow{Ether} \left[CH_3CH_2-\underset{H}{\overset{CH_3}{\underset{|}{\overset{|}{C}}}}-OMgBr\right] \xrightarrow[H^+]{H_2O} CH_3CH_2\underset{OH}{\underset{|}{C}}HCH_3 + Mg(OH)Br$

Ethyl magnesium bromide — Ethanal (Acetaldehyde) — Butan-2-ol

$CH_3(CH_2)_3MgBr + CH_3COCH_3 \xrightarrow{Ether} \left[CH_3(CH_2)_3-\underset{CH_3}{\overset{CH_3}{\underset{|}{\overset{|}{C}}}}-OMgBr\right] \xrightarrow[H^+]{H_2O} CH_3(CH_2)_3-\underset{CH_3}{\overset{CH_3}{\underset{|}{\overset{|}{C}}}}-OH + Mg(OH)Br$

n-butyl magnesium bromide — Propanone (Acetone) — 2-methyl hexan-2-ol

Formate ester is used to prepare a secondary alcohol with identical alkyl groups

Example

$2CH_3MgBr + H\text{-}\overset{O}{\overset{\|}{C}}\text{-}O\text{-}CH_2\text{-}CH_3 \xrightarrow[ii)\ H_3O^+]{i)\ \text{ether solvent}} CH_3-\underset{OH}{\underset{|}{C}}H-CH_3 + Mg\begin{matrix}Br\\OCH_2\text{-}CH_3\end{matrix}$

ethylmethanoate — propan - 2 - ol

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